Table of Content

Page 1	Title Page
Page 2	BLOCK Diagram
Page 3	Power Tree
Page 4	CLK Tree
Page 5	change
Page 6	Hub Connector
Page 7	FPGA
Page 8	FPGA_PWR
Page 9	DDR3
Page 10	GigaPhy
Page 11	ETH Port
Page 12	POWER

PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

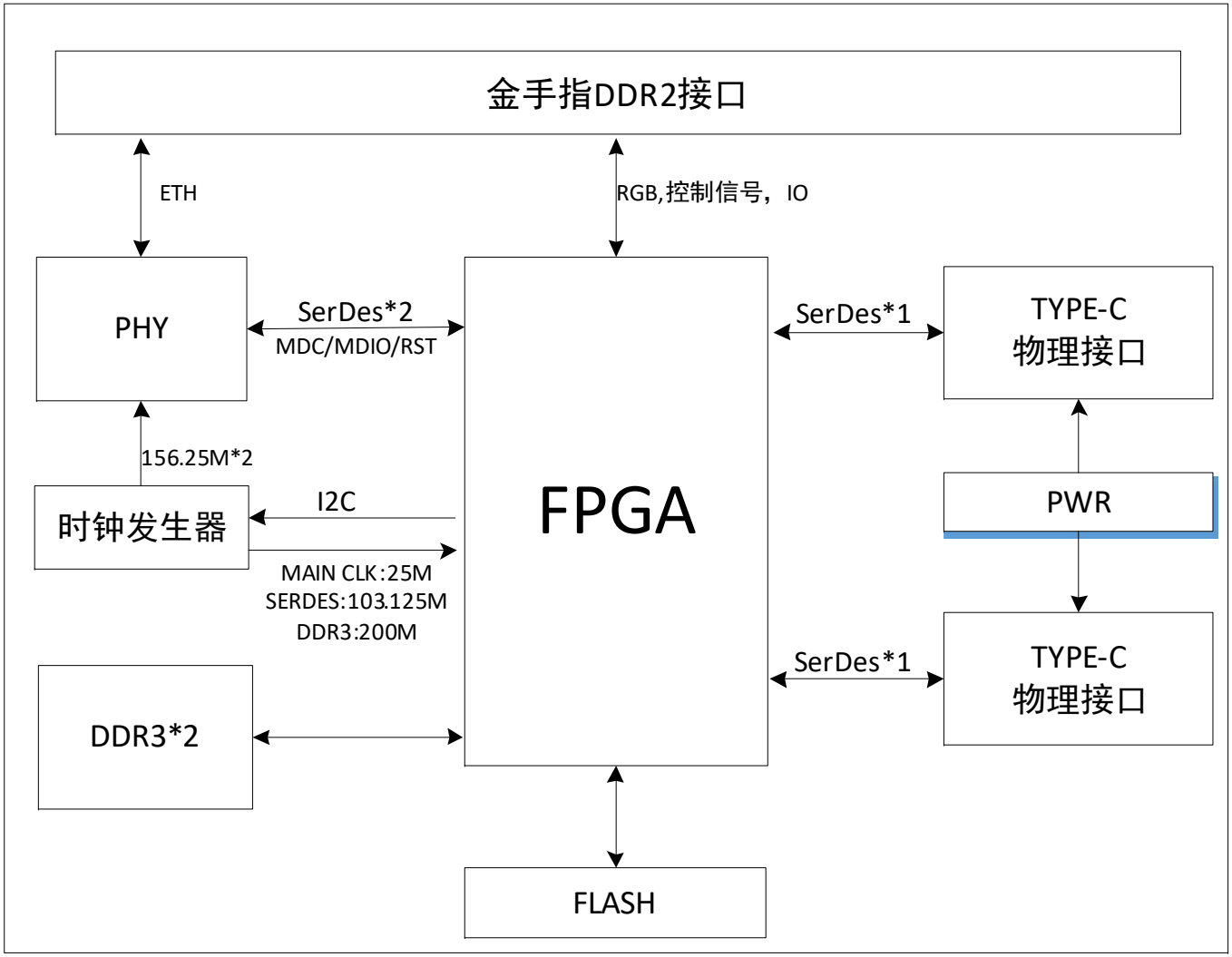
STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

Xi'an Novastar Technologies,Inc.

http://www.novastar-led.cn TEL:+86-029-68216000

Title:	01-title page		
Size: A4	Project: X50C_A0	Document Number: <Doc2>	Rev: <Rev>
Designer: WX	CHKD: <Name2>	SUPV: <Name3>	
Date: Monday, July 12, 2021	Sheet: 1 of 11		



PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

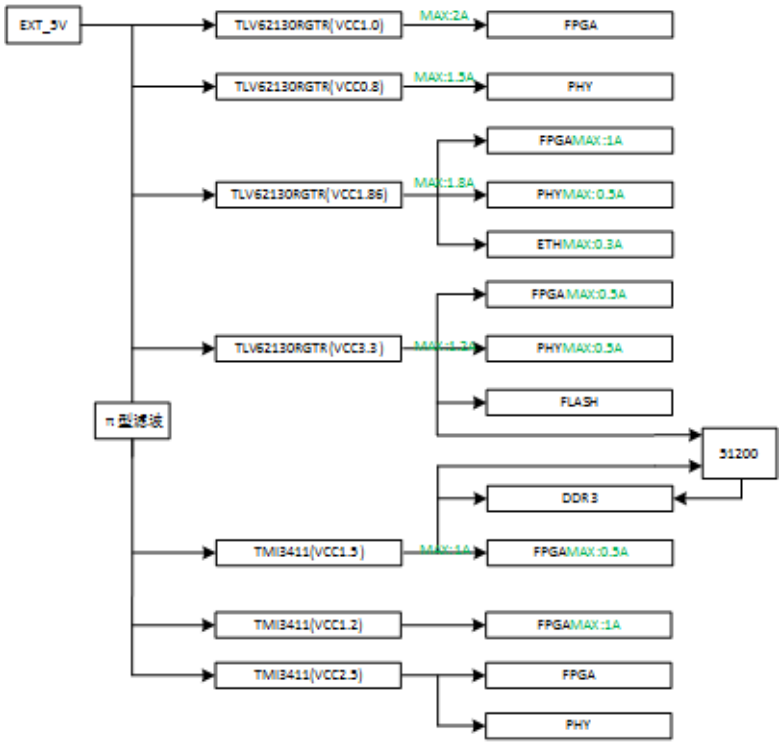
NOVASTAR 诺瓦科技 Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

Title: **02.Block Diagram**

Size	Project:	Document Number:	Rev:
A4	X50C_A0	<Doc2>	<Rev>

Designer:	CHKD:	SUPV:
WX	<Name2>	<Name3>

Date: Monday, July 12, 2021 Sheet: 2 of 11



PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

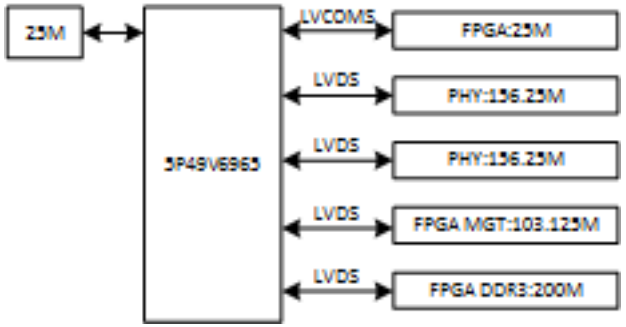
Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

Title: **03-Power Tree**

Size	Project:	Document Number:	Rev:
A4	X50C_A0	<Doc2>	<Rev>

Designer:	CHKD:	SUPV:
WX	<Name2>	<Name3>

Date: Monday, July 12, 2021 Sheet: 3 of 11



PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

Title: **04-Clock Tree**

Size	Project:	Document Number:	Rev:
A4	X50C_A0	<Doc2>	<Rev>

Designer:	CHKD:	SUPV:
WX	<Name2>	<Name3>

Date: Monday, July 12, 2021 Sheet: 4 of 11

S50C_D0	
S50C_E0	添加一路TYPE-C接口，每个type-c链接一组serdes； PCB结构更改；
X50C_A0	将高密接插件更改为金手指DDR2接口
X50C_B0	详见改版变更清单

PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

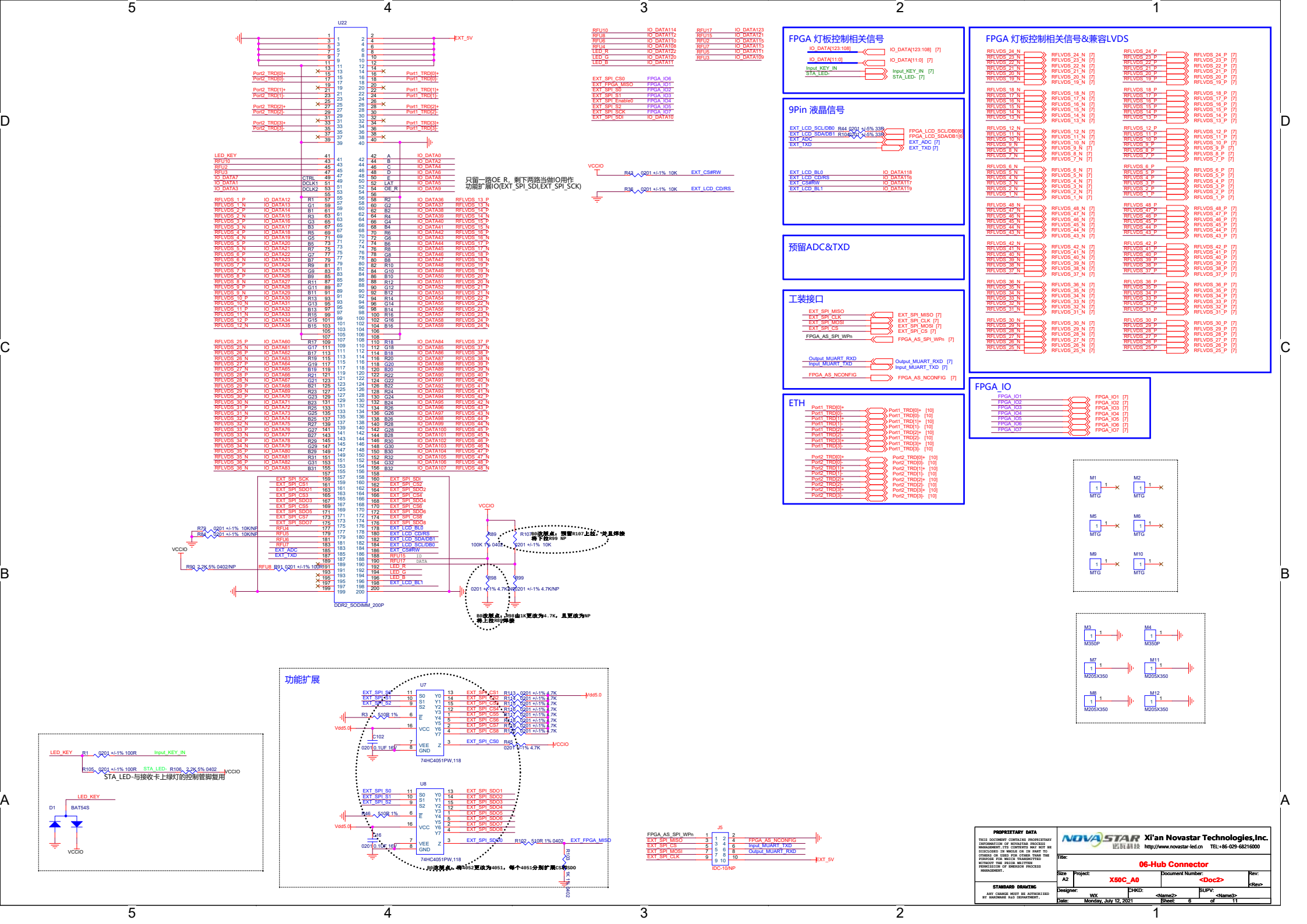
NOVASTAR 诺瓦科技 Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

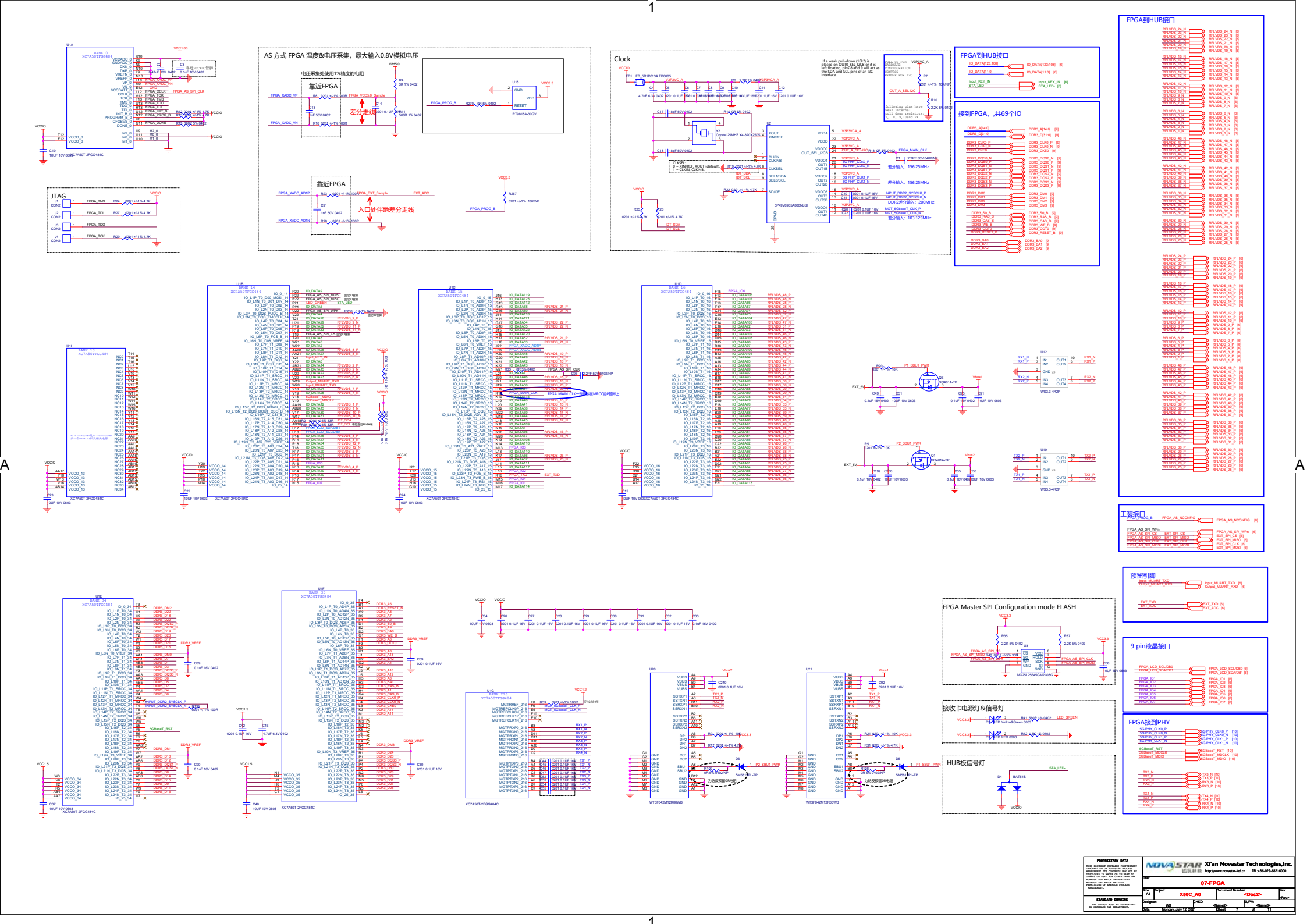
Title: 05-Change

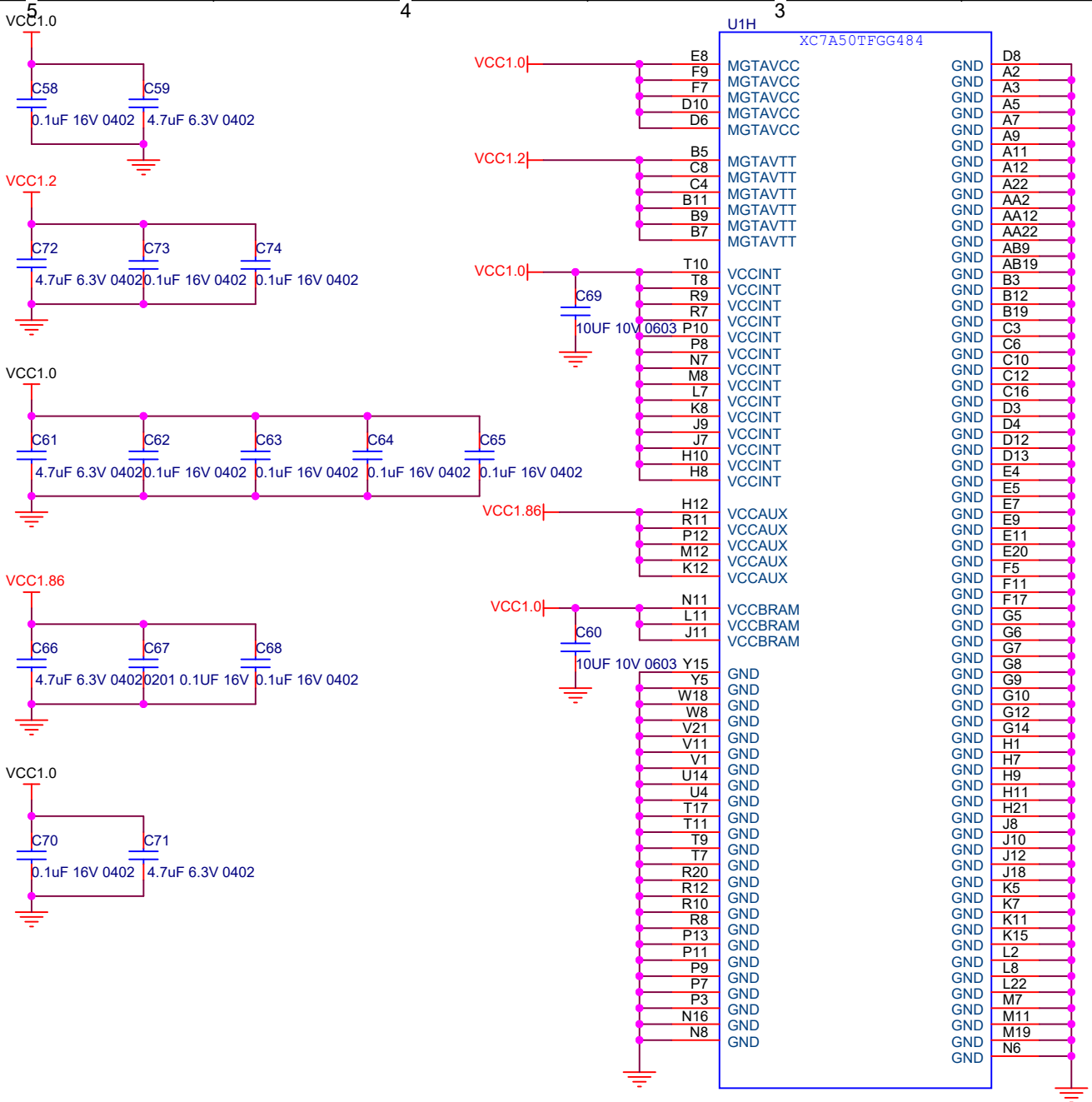
Size	Project:	Document Number:	Rev:
A4	X50C_A0	<Doc2>	<Rev>

Designer:	CHKD:	SUPV:
WX	<Name2>	<Name3>

Date: Monday, July 12, 2021 Sheet: 5 of 11


FPGA 灯板控制相关信号
9Pin 液晶信号
预留ADC&TXD
工装接口
ETH
FPGA 灯板控制相关信号&兼容LVDS
FPGA_IO
只留一路OE_R，剩下两路当做IO用作功能扩展IO(EXT_SPI_SDI,EXT_SPI_SCK)
功能扩展
STA_LED-与接收卡上绿灯的控制管脚复用
DDR2_SODIMM_200P
74HC4051PW,118
IDC-10/NP
PROPRIETARY DATA
STANDARD DRAWING
Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000
06-Hub Connector
X50C_A0
Monday, July 12, 2021


AS 方式 FPGA 温度&电压采集，最大输入0.8V模拟电压
电压采集处使用1%精度的电阻
靠近FPGA
差分走线
靠近FPGA
入口处伴地差分走线
Clock
FPGA到HUB接口
接到FPGA，共69个IO
JTAG
FPGA到HUB接口
工装接口
预留引脚
9 pin液晶接口
FPGA Master SPI Configuration mode FLASH
接收卡电源灯&信号灯
HUB板信号灯
FPGA接到PHY
XC7A50TFGG484C
XC7A50T-2FGG484C
WT3F042M12R00WB
SMSB1ML-TP
BAT54S
Q3401A-TP
WS3.3-4R2P
MX25L2564GM2I-08G
07-FPGA
X58C_A0
XI'an Novastar Technologies,Inc.
http://www.novastar-led.cn
TEL:+86-029-68216000



PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

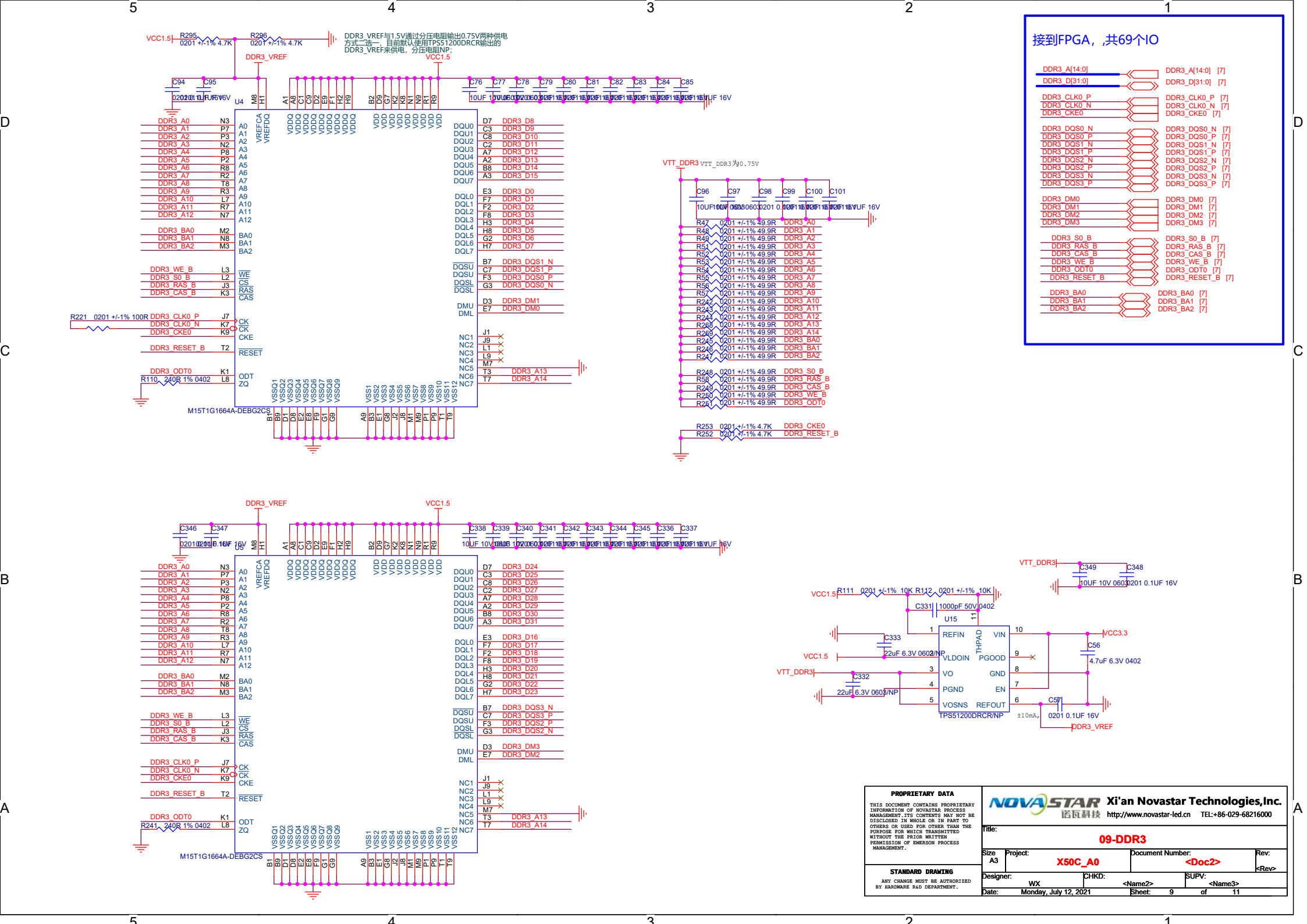
Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

Title: **08-FPGA-PWR**

Size	Project:	Document Number:	Rev:
A4	X50C_A0	<Doc2>	<Rev>

Designer:	CHKD:	SUPV:
WX	<Name2>	<Name3>

Date: Monday, July 12, 2021 Sheet: 8 of 11



PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT. ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

Title: 09-DDR3

Size	Project	Document Number	Rev
A3	X50C_A0	<Doc2>	<Rev>

Designer	CHKD	SUPV
WX	<Name2>	<Name3>

Date: Monday, July 12, 2021 Sheet: 9 of 11

U6A

BCM54992B0KFSBG

When high, it selects 156.25 MHz for the reference clock frequency.

U6B

BCM54992B0KFSBG

差分输入：156.25MHz

5GBaseT_INT0 R63 0201 +/-1% 4.7K/NP VCC1.86

5GBaseT_INT1 R64 0201 +/-1% 4.7K/NP

用来选用是否加载SPI FLASH：
上拉：选用加载FLASH

U6D

BCM54992B0KFSBG

U6C

BCM54992B0KFSBG

U6E

BCM54992B0KFSBG

R0改版点：将电容需要更改为X7R

RDAC Bias Resistor. Adjusts the reference current of the transmitter DAC. A 6.04 kΩ ±1% resistor is connected between RDAC0 and GND for port 0, and between RDAC1 and GND for port 1.

Resistor CAL for ports 0 and 1. Place a 4.53 kΩ ±1% between RCALP and GND.

JTAG Reset. Active-low. Resets the JTAG controller. This signal must be pulled low during normal operation.

PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT. ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

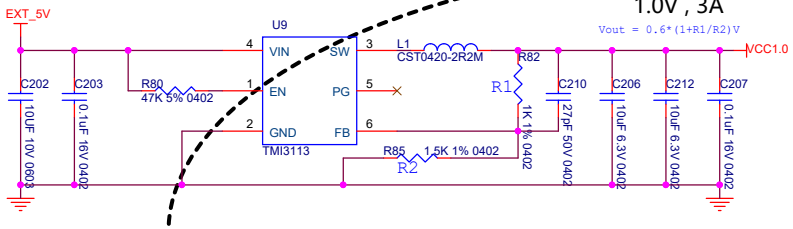
Title:	10-GigaPhy		
Size: A3	Project: X50C_A0	Document Number: <Doc2>	Rev: <Rev>
Designer: WX	CHKD: <Name2>	SUPV: <Name3>	
Date: Monday, July 12, 2021	Sheet: 10 of 11		

不考虑FPGA上电时序

电源处使用1%精度的电阻

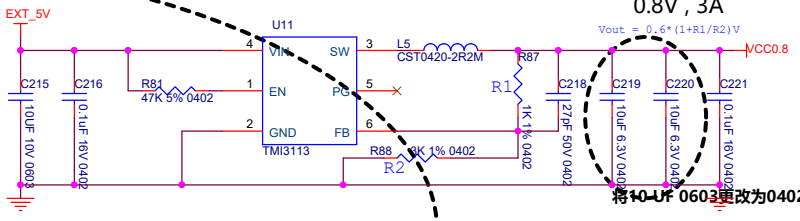
电感用一体成型的CST0420-2R2M:温升电流3A

1.0V , 3A



电感用一体成型的CST0420-2R2M:温升电流3A

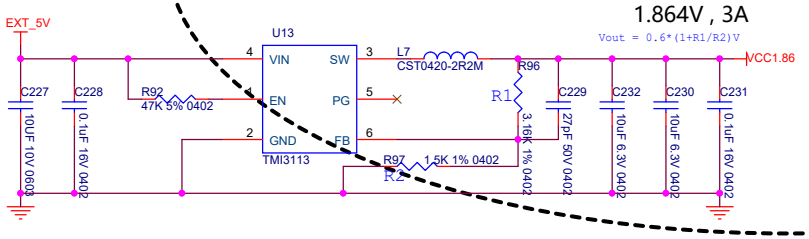
0.8V , 3A



将10uF 0603更改为0402封装

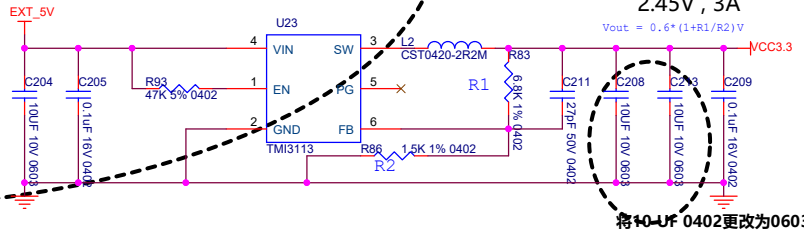
将62130A更改为TMI3113

1.864V , 3A

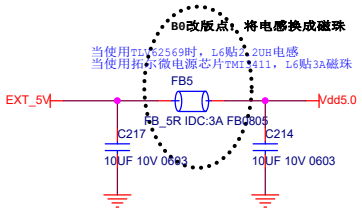


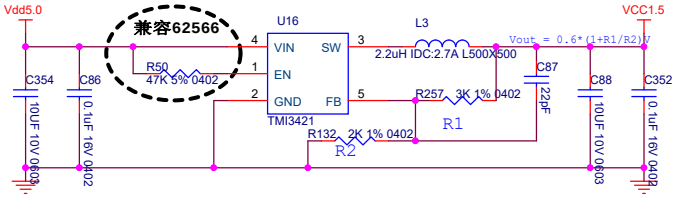
电感用一体成型的CST0420-2R2M:温升电流3A

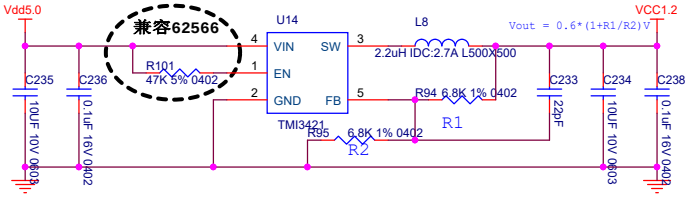
2.45V , 3A

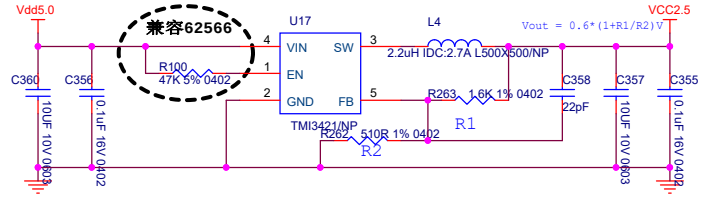


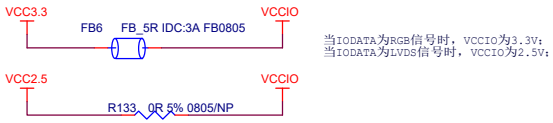
将10uF 0402更改为0603封装











当IODATA为RGB信号时，VCCIO为3.3V；
当IODATA为LVDS信号时，VCCIO为2.5V；

PROPRIETARY DATA

THIS DOCUMENT CONTAINS PROPRIETARY INFORMATION OF NOVASTAR PROCESS MANAGEMENT.ITS CONTENTS MAY NOT BE DISCLOSED IN WHOLE OR IN PART TO OTHERS OR USED FOR OTHER THAN THE PURPOSE FOR WHICH TRANSMITTED WITHOUT THE PRIOR WRITTEN PERMISSION OF EMERSON PROCESS MANAGEMENT.

STANDARD DRAWING

ANY CHANGE MUST BE AUTHORIZED BY HARDWARE R&D DEPARTMENT.

NOVASTAR 诺瓦科技 Xi'an Novastar Technologies,Inc.
http://www.novastar-led.cn TEL:+86-029-68216000

Title: 11-POWER

Size A3 | Project: X50C_A0 | Document Number: <Doc2> | Rev: <Rev>

Designer: WX | CHKD: <Name2> | SUPV: <Name3>

Date: Monday, July 12, 2021 | Sheet: 11 of 11